

Question Paper

Exam Date & Time: 26-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Operating Systems [IT_3128]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Apply your knowledge of the storage hierarchy to explain the trade-off between speed, cost, and volatility at different levels of memory. (4)
- A) Develop an example scenario where caching improves system performance significantly, and discuss what would happen if cache management is poorly implemented.

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Develop a detailed process tree representing a parent process that creates three child processes, where: (4)
- The first child executes concurrently with the parent,
 - The second shares some resources, and
 - The third runs only after the parent terminates.

Explain how the operating system manages resource sharing, execution synchronization, and termination for these processes.

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Consider the following code snippet: (2)
- ```
#include <stdio.h>
#include <unistd.h>
```

```
int main() {
 pid_t pid = fork();
 if (pid == 0) {
 execlp("ls", "ls", NULL);
 printf("This line will not execute");
 } else {
 printf("Parent process executing");
 }
 return 0;
}
```

It is observed that the following statement  
printf("This line will not execute");  
is never printed by the child process.  
Demonstrate the reason for this behaviour of the code snippet.

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) A system uses preemptive priority scheduling. The priority values are integers where a smaller (4)

value = higher priority.

- A) At time  $t = 0$ , three processes arrive:  
P1(3 ms, priority 3), P2(4 ms, priority 2), P3(5 ms, priority 1).  
Aging increases a process's priority by 1 every 3 ms of waiting.  
Examine how the aging mechanism affects the execution order if the system becomes heavily loaded.  
Provide a reasoned inference on whether starvation can still occur under extreme process arrival rates.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Consider a system implementing the first readers-writers problem using semaphores: (4)  
Semaphore wrt = 1, mutex = 1;  
int readcount = 0;  
The code structure is as follows:  
Reader:  
wait(mutex);  
readcount++;  
if (readcount == 1)  
wait(wrt);  
signal(mutex);  
// reading  
wait(mutex);  
readcount--;  
if (readcount == 0)  
signal(wrt);  
signal(mutex);  
  
Writer:  
wait(wrt);  
// writing  
signal(wrt);  
Suppose three readers (R1, R2, R3) and one writer (W1) arrive in this order:  
R1 → R2 → W1 → R3.  
Apply the algorithm to determine the sequence of execution that satisfies all semaphore conditions.  
Identify whether starvation can occur in this case and justify your answer mathematically using semaphore values.

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) A concurrent system allows multiple processes to execute while sharing common resources. (2)  
Analyze the essential requirements that any synchronization mechanism must fulfill to ensure correct and fair access to a shared resource.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3) A system contains 3 processes (P1, P2, P3) and a single resource type R with 12 total instances. (4)  
At a particular time, the allocation and maximum needs are:

A)

| Process | Allocation | Maximum |
|---------|------------|---------|
| P1      | 5          | 10      |
| P2      | 2          | 4       |
| P3      | 3          | X       |

Available = 2 instances

- Make use of the concept of safe state to solve for the maximum possible value of X for which the system remains safe.
- Utilize the Banker's safety condition and plan your steps clearly.

- Identify if increasing X beyond this limit will lead to a potential deadlock.

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Two threads T1 and T2 execute concurrently. (4)
- T1 locks resource L1 and then requests L2.
  - T2 locks resource L2 and then requests L1.

Assume both threads hold their first locks simultaneously before requesting the second.  
Model this situation using the four necessary conditions for deadlock and identify which of them are satisfied.  
Then, develop a strategy that the system designer can apply to prevent such a deadlock, while ensuring minimal impact on concurrency.

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) A company designs a new OS for embedded automotive systems. The design team focuses mainly on reliability, efficiency, and resource management, while users expect only smooth functionality. Compare the user view and system view of the operating system in this scenario, to understand their influences on the OS design. (2)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 4) Assume a program is compiled, loaded, and executed on a system supporting call 3 types of binding. (4)
- A) Identify how the memory address of a variable changes under each binding type. Support your explanation with an example illustrating how a single variable's address may differ under each scheme.

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) Consider a system with 500 KB of free memory divided into holes of sizes 200 KB, 150 KB, 100 KB, and 50 KB. A new process of size 120 KB arrives. (4)
- Model how this process will be allocated under First-Fit, Best-Fit, and Worst-Fit strategies. Identify which method leads to minimum external fragmentation and justify mathematically.

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) A newly designed system implements pure demand paging for all processes. During testing, engineers notice that some processes experience excessive page faults immediately after starting, even though the total memory available is sufficient. (2)
- Examine this situation and explain how the principle of locality of reference helps in reducing the number of initial page faults.  
Suggest one modification in the paging policy that could minimize this startup behaviour.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 5) A process references the following pages in order: (4)
- 1, 2, 3, 4, 2, 1, 5, 2, 4, 3, 2, 1, 6, 7, 3, 2, 1.
- A) The system has 3 page frames.  
Distinguish the number of page faults generated by the Optimal Page Replacement and LRU algorithms.  
Construct the page frame table for both methods and draw a short inference about which algorithm minimizes replacement under this workload.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) A disk queue contains requests for cylinders in the order: (4)
- 95, 180, 34, 119, 11, 123, 62, 64

The disk head is currently positioned at cylinder 50, and the lowest and highest cylinder numbers on the disk are 0 and 199, respectively.  
 Apply the SSTF and FCFS disk scheduling algorithms to this request sequence.  
 Compute the total head movement for both algorithms and identify which one yields better average seek performance.  
 Show your calculation and reasoning step by step.

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Different directory structures (single-level, two-level, tree, acyclic graph) are used in modern operating systems. (2)  
 Construct a suitable directory organization model for a system used by multiple users who frequently share project files but also maintain private workspaces.

Course Outcome: "CO4" / Cognitive Level: "Application"

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| Event Name | Subject |
|------------|---------|
|------------|---------|

058-END SEMSTER-ODD-2025 IT\_3128:Operating Systems

| #            | Course Outcome | Description | Marks     |
|--------------|----------------|-------------|-----------|
| 1            | 1              | CO 1        | 12        |
| 2            | 2              | CO 2        | 18        |
| 3            | 3              | CO 3        | 18        |
| 4            | 4              | CO 4        | 2         |
| <b>Total</b> |                |             | <b>50</b> |

\* **NA** - CO Not Selected

| Event Name | Subject |
|------------|---------|
|------------|---------|

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| #            | Level       | No of Questions | Marks (%)        |
|--------------|-------------|-----------------|------------------|
| 1            | Remember    | 0               | 0(0%)            |
| 2            | Understand  | 0               | 0(0%)            |
| 3            | Application | 10              | 36(72%)          |
| 4            | Analysis    | 5               | 14(28%)          |
| 5            | Create      | 0               | 0(0%)            |
| 6            | Evaluate    | 0               | 0(0%)            |
| <b>Total</b> |             | <b>15</b>       | <b>50 (100%)</b> |